

CSE 3203 CT 4 Assignment

Roll No: 1903002

Assignment Problem:

Build CPU based on following requirements:

1. Word Size of CPU = 3
2. ALU Operations = NOT, ADD, SHR
3. Register Number = 3
4. Size of RAM = 9
5. Word size of ISA and RAM = 17
6. CPU Instructions = Register Mode, Immediate Mode, JMP, JC

Solution:

Video Link (Youtube Video Link/Google Drive Link with Necessary Drive Permissions):

https://drive.google.com/file/d/1tfSyymbtcWQVjKZOuewx_z5uOzwxnaAr/view?usp=sharing

Simulator Design:

1. ALU Circuit (Show all circuits except FA circuit)(Marks 5):

Check List:

Have you added all circuits of ALU from FA to ALU Operations Circuits (ADD, XOR, SHL etc.) to Top Level ALU Circuit?
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YES

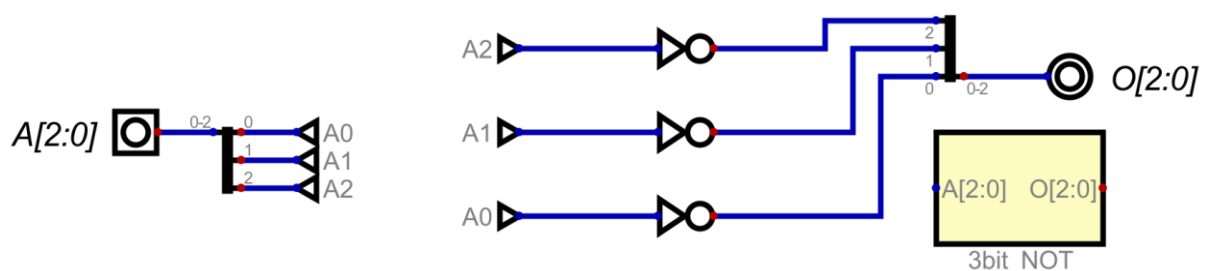


Figure: 3bit NOT

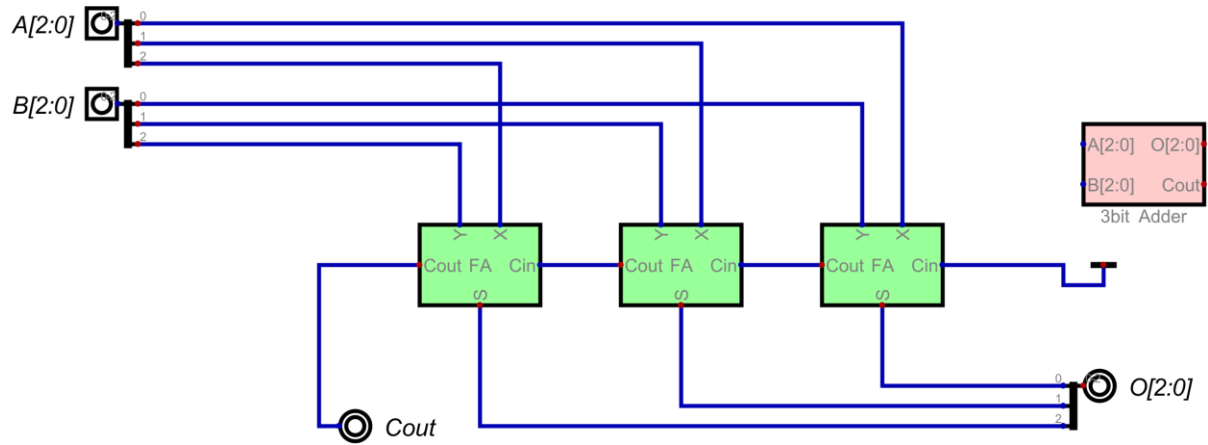


Figure: 3bit Adder

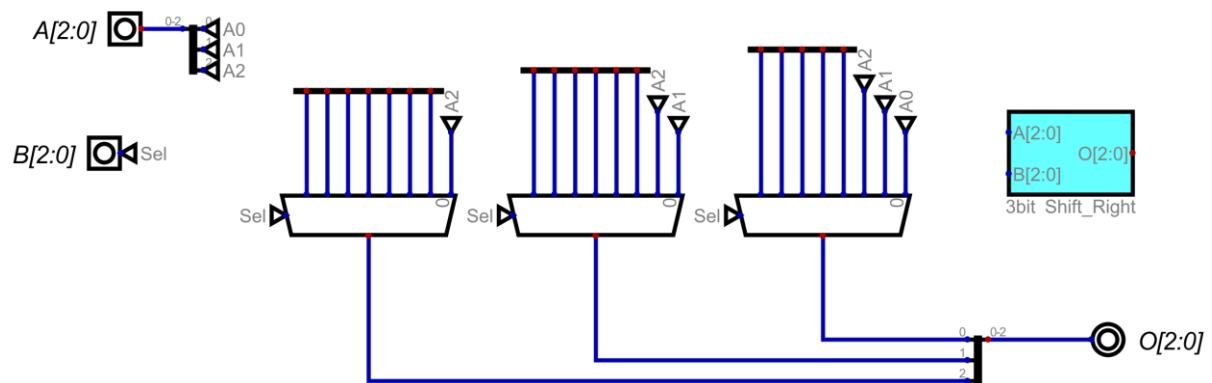


Figure: 3bit Shift Right

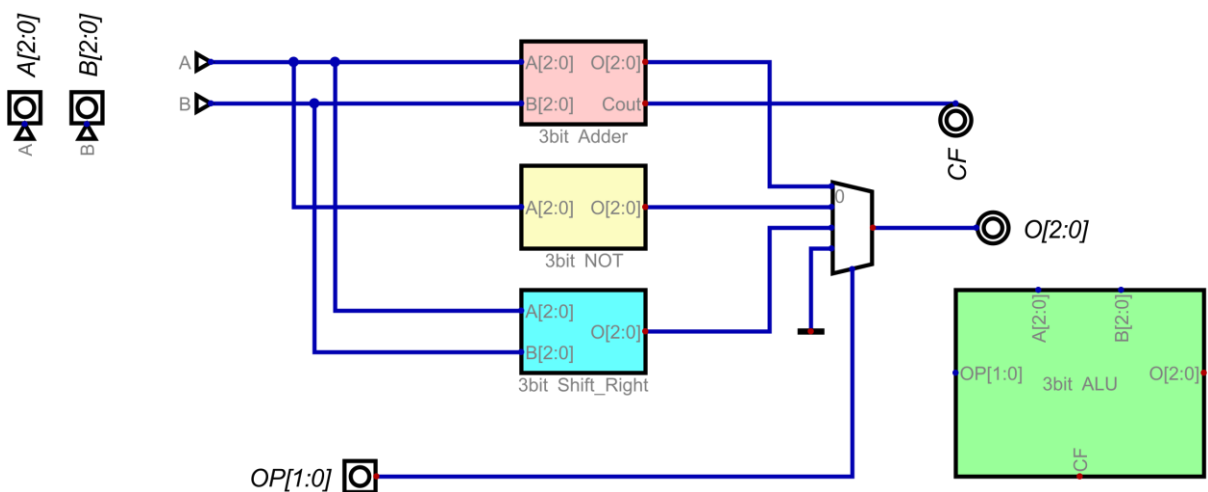


Figure: 3bit ALU

2. Register Set Circuit (Top to Bottom all circuits)(5 Marks):

Check List:

Have you added all circuits of Register Set from 1 bit Register to n bit Register to Top Level Register Set Circuit.?

YES

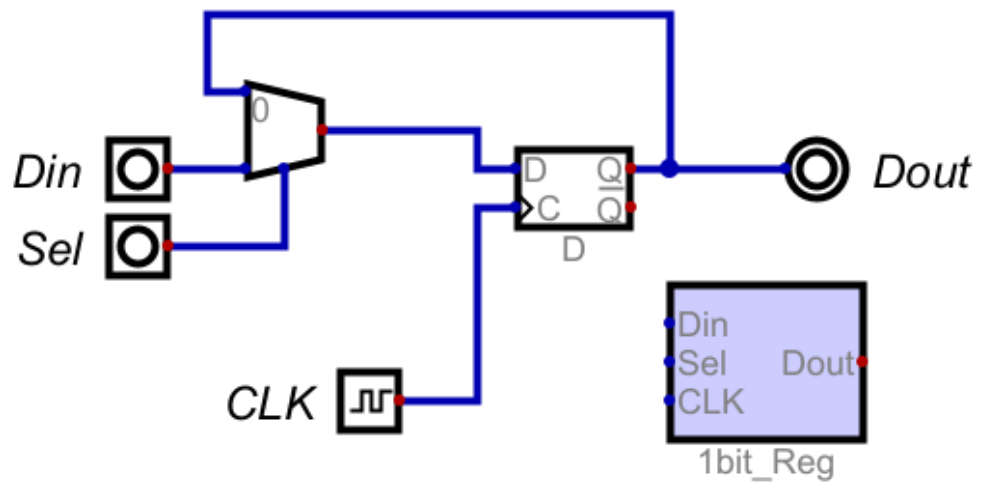


Figure: 1bit Register

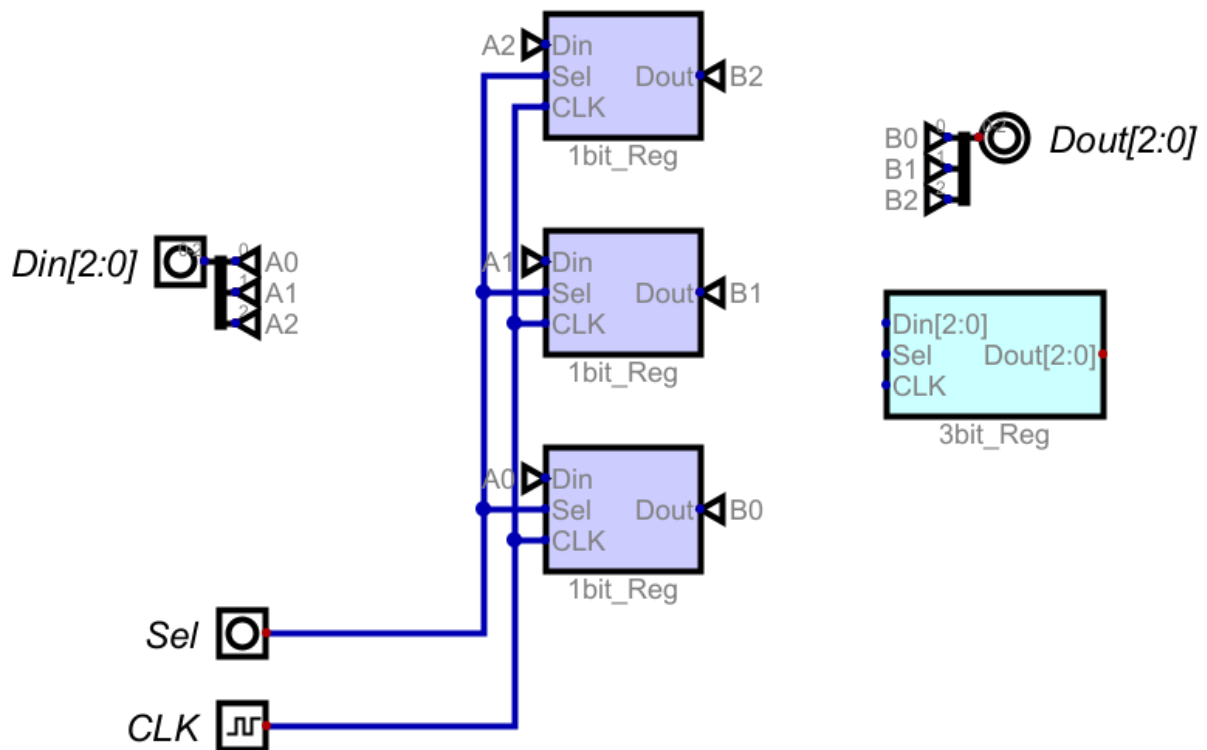


Figure: 3-bit Register

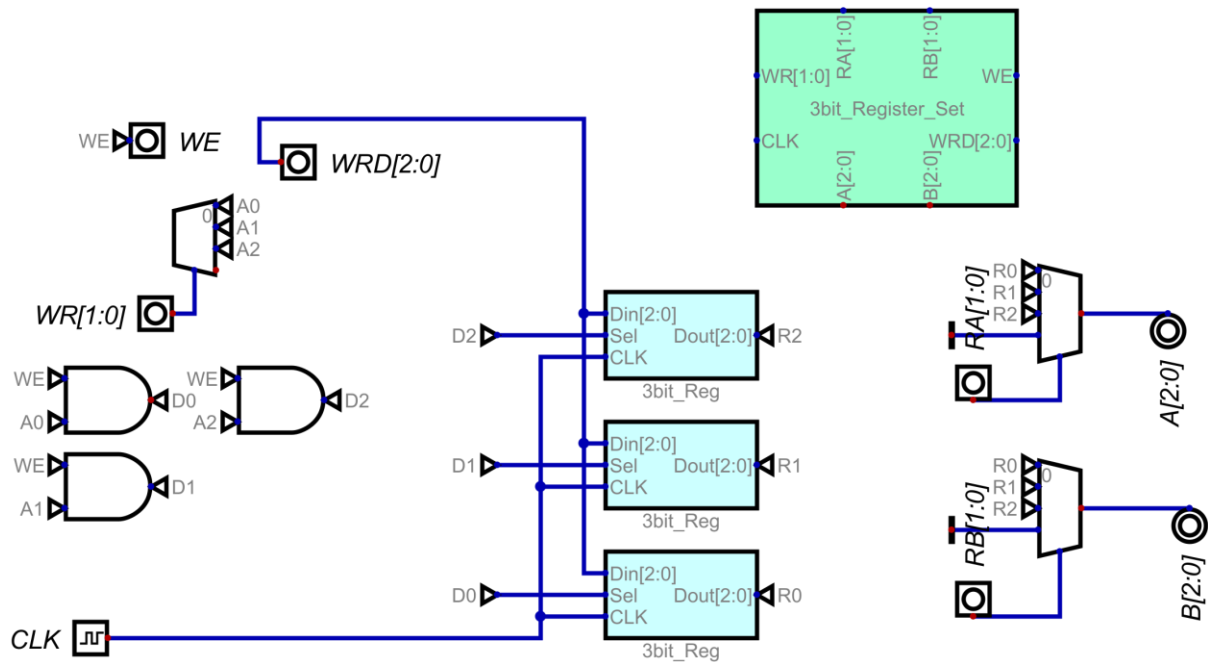


Figure: 3-bit Register Set with 3 Registers Chip

3. RAM Circuit (Top to Bottom all circuits)(5 Marks):

Check List:

Have you added all circuits of RAM from 1x1 RAM to 1xN RAM to MxN RAM?	YES
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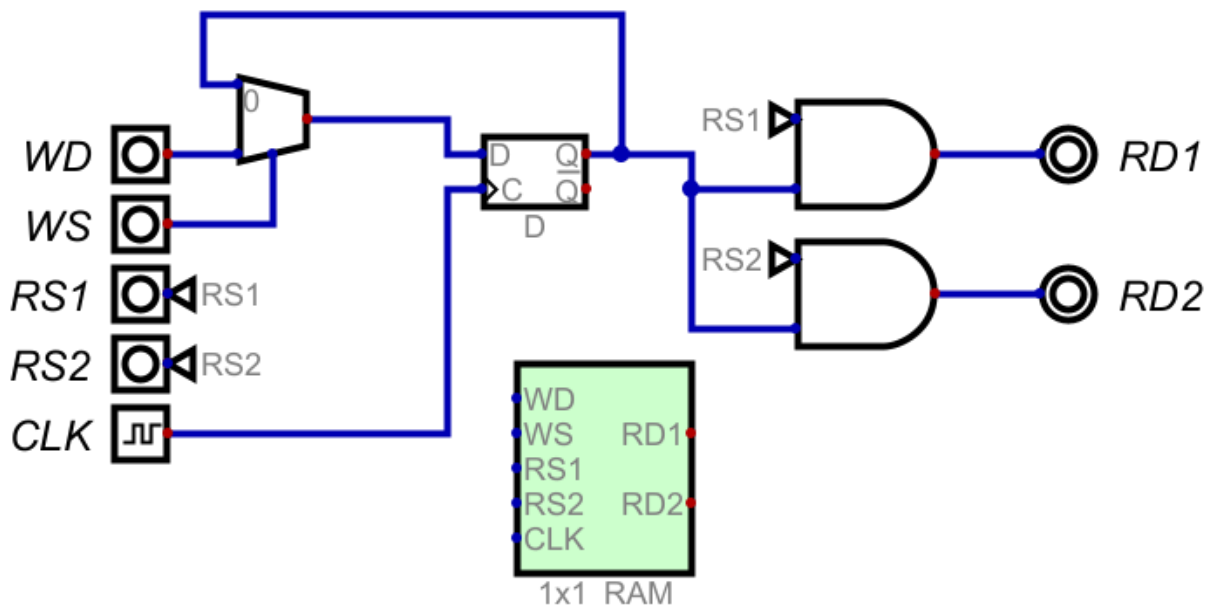


Figure: 1X1 RAM

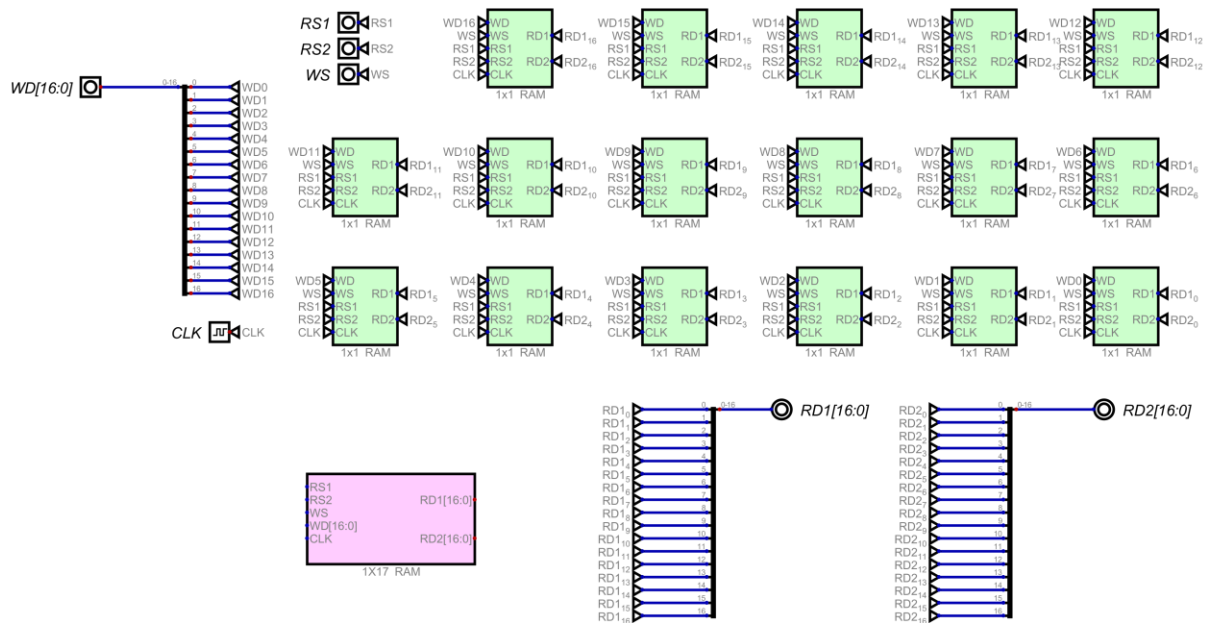


Figure: 1X17 RAM

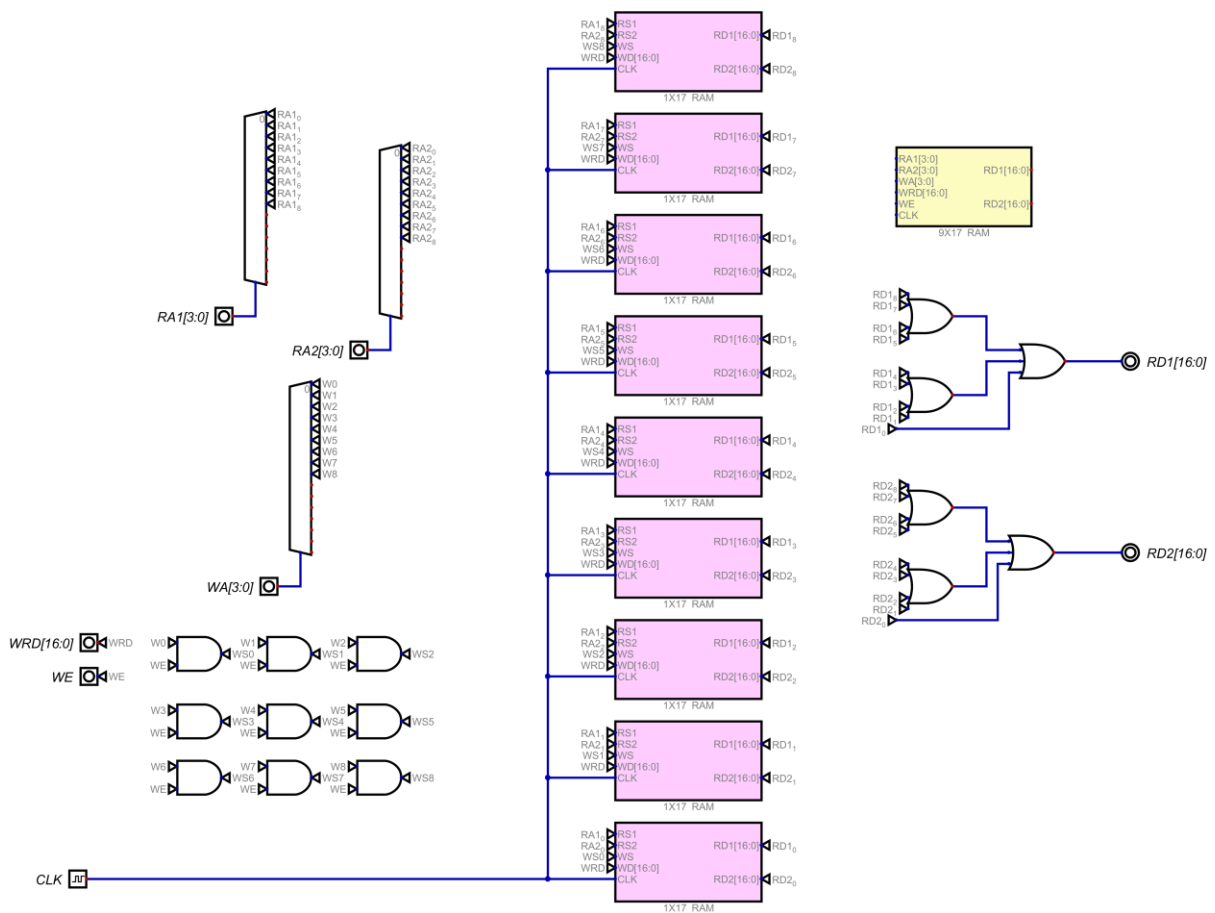


Figure: 9X17 RAM

4. a) ISA (2 Marks)

Check List:

Have you added all ISA of CPU along with its sample machine code to be run on CPU?	YES
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ISA (Register Mode):

Opcode (4 bit)		Register 1	Register 2	Unused
2 bits	2 bits	2 bits	2 bits	9 bits
Types of instruction=00	Operations (ADD=00, NOT=01, SHR=10)	Ra (00-10)	Rb (00-10)	X

ISA (Immediate Mode):

Opcode (4 bit)		Register 1	Constant	Unused
2 bits	2 bits	2 bits	3 bits	8 bits
Types of instruction=01	Operations (ADD=00, NOT=01, SHR=10)	Ra (00-10)	Value(000-111)	X

ISA (Branching):

Opcode (4 bit)		Address	Unused
2 bits	2 bits	4 bits	9 bits
Types of instruction=10	Operations (JMP=00, JC=01)	Value (0000-1111)	X

Sample Machine Code with assembly code in comments to be run on CPU (You will make a video running this machine code on CPU in order to prove that your CPU is working perfectly)

Machine Code	Assembly Code
10000001000000000	JMP START
01000010100000000	START: ADD R1,5
01000010000000000	JMPTO:ADD R1,4

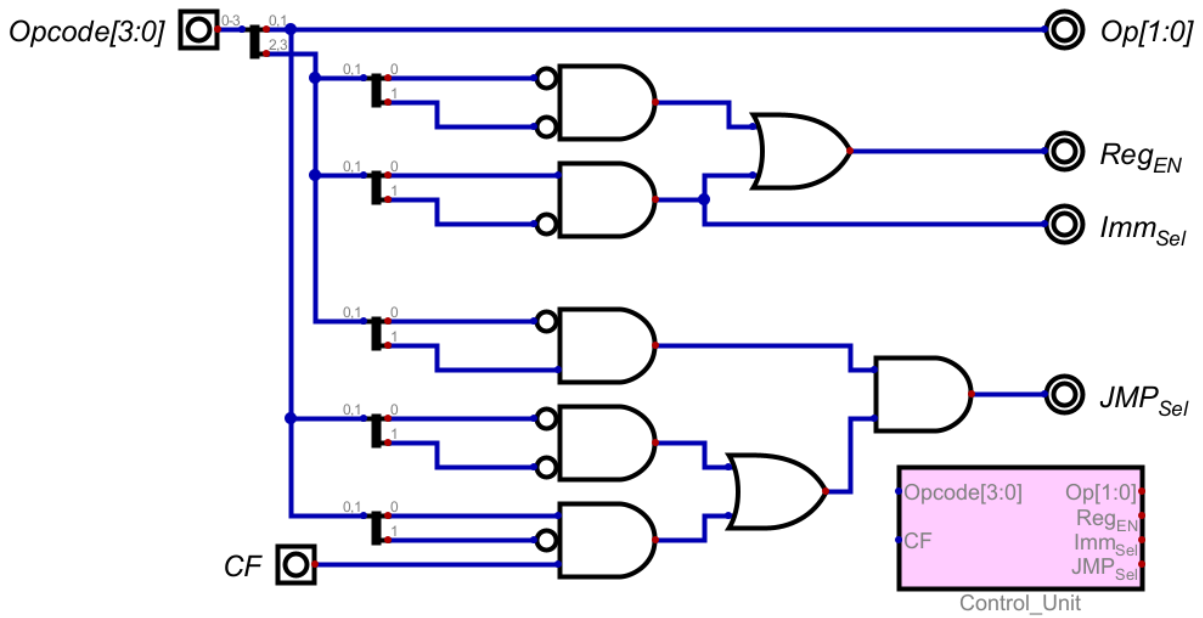


Figure: Control Unit

Assembly :

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JMP START->10000001000000000
START: ADD R1,5->01000010100000000
JMPTO:ADD R1,4->01000010000000000
JC JMPTO->10010010000000000
ADD R2,1->01000100100000000
SHR R1,R2->00100001000000000
NOT R1->00010000000000000
  
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Register Mode(type of Op=00)= 2bit(Type of Op)+ 2bit(Op)+2bit(Reg1)+2bit(Reg2)+9bit(don't care)
 Immediate Mode(type of Op=01)= 2bit(Type of Op)+ 2bit(Op)+2bit(Reg1)+3bit(Imm value)+8bit(don't care)
 Branching(type of Op=10)= 2bit(Type of Op)+ 2bit(Op)+4bit(JMP Add)+9bit(don't care)
 ADD(Op=00), NOT(Op=01), SHR(Op=10)
 JMP(Op=00),JC(01)

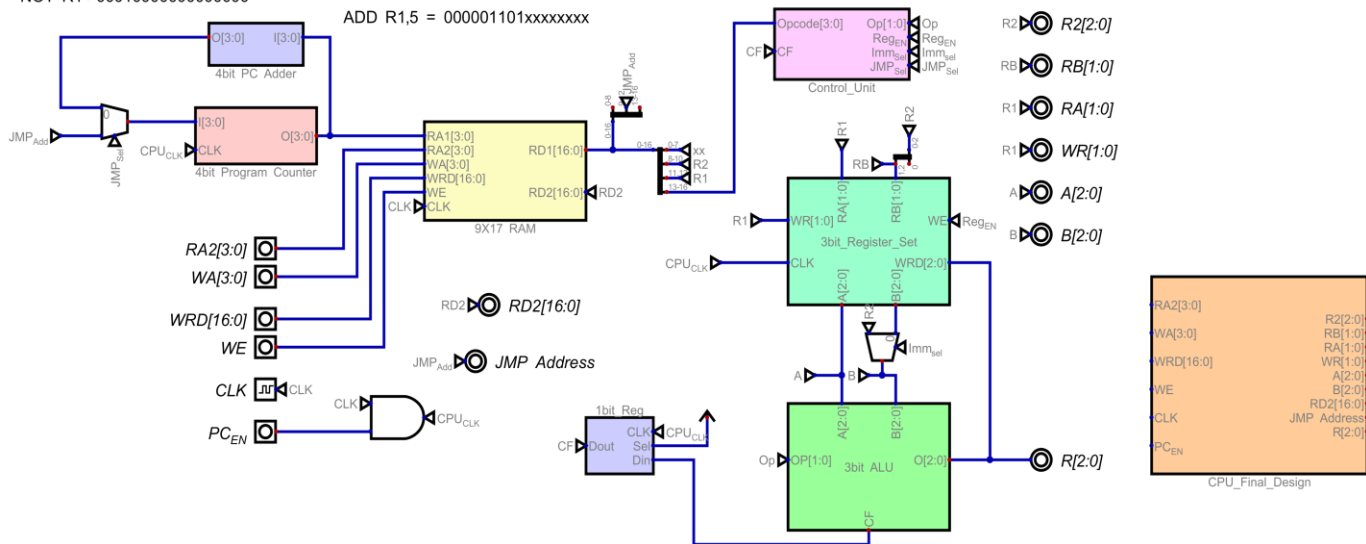


Figure: Final CPU